

# Technical Design & Architecture Document

## SALESFORCE DATA CLOUD SEMANTIC MODELING & AGENTFORCE IMPLEMENTATION SPECIFICATION

## 1. Executive Summary & Data Ingestion Pipeline

This architectural specification details the data lineage, algorithmic configuration, and deployment topology required to ingest customer master records and high-velocity storefront clickstream behavioral telemetry. The implementation maps disparate tables across Salesforce Data Cloud, structures quantitative engagement scoring frameworks, and registers an actionable context matrix to ground an autonomous Agentforce Advisory Subagent pipeline.

The foundation layer streams and ingests customer profile arrays from two source environments directly into Data Lake Objects (DLOs) inside Salesforce Data Cloud:

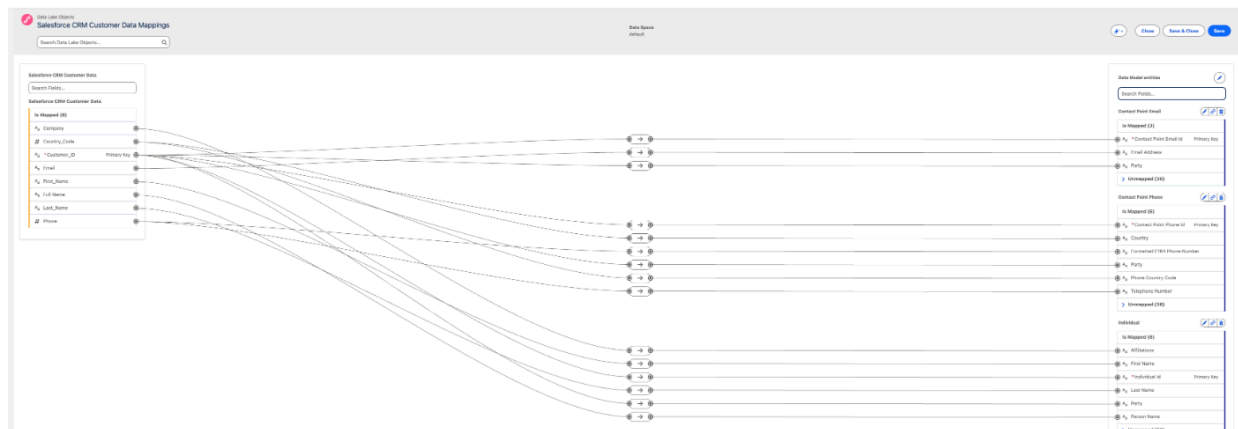
- **Salesforce CRM Customer Data Stream:** Captures master registration profiles containing physical demographics, baseline account cross-references, telephone lines, and primary subscription markers.
- **Web Engagement Data Stream (Myntra Storefront):** Ingests low-latency client-side interaction events including page views, click volumes, items added to active wishlists, shopping baskets, cart updates, and current session authorization status tracking (Known vs. Unknown users).

## 2. Canonical Semantic Modeling & Infrastructure Modification

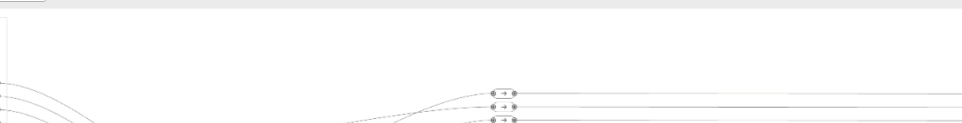
To bridge behavioral tracking with CRM master records into a single harmonized identity format, properties from ingested DLO data structures are mapped systematically using the standard Data Cloud Mapping Engine into primary canonical target Data Model Objects (DMOs):

- **Individual DMO:** Acts as the centralized semantic entity mapping layer for primary personal attributes, unifying cross-device session keys under a coordinated master customer record track.
- **Contact Point Email DMO:** Normalizes electronic communication channels, standardizing variable string parameters to successfully detect and resolve duplicate profiles.
- **Contact Point Phone DMO:** Structures customer phone connection metrics, routing raw scientific phone floats into reliable backend target tracks.

## Salesforce CRM Customer Data Mappings



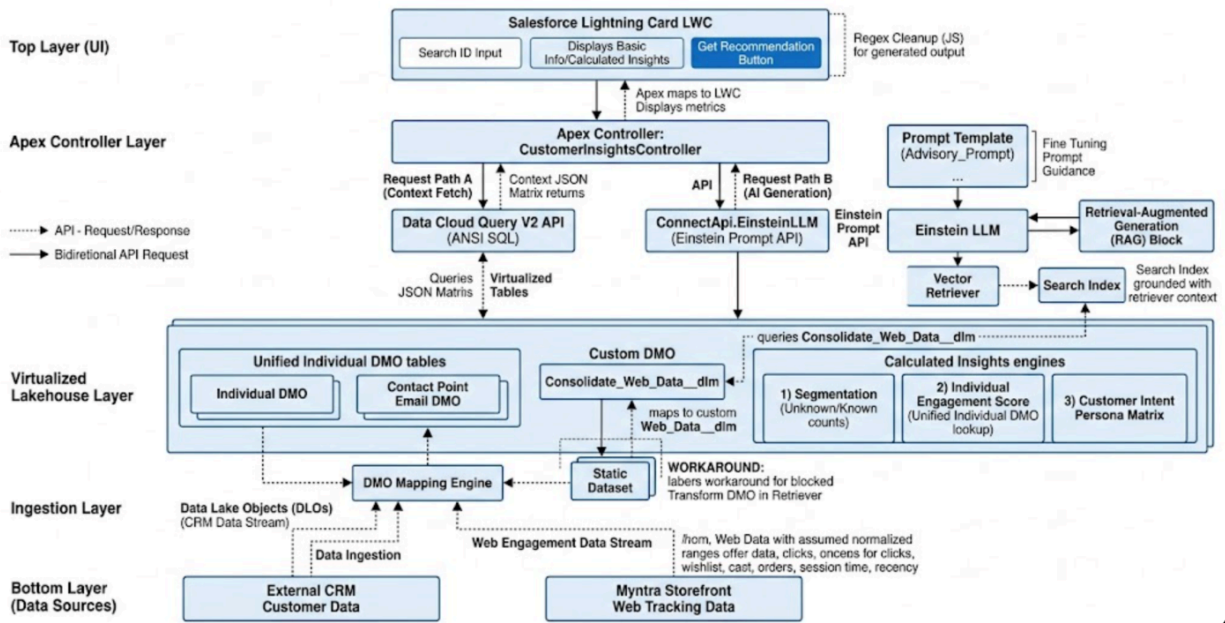
## Web Engagement Data From Myntra Mappings



The screenshot displays the Mynta Mapper tool interface, which is used for creating and managing data models. The interface is divided into three main sections:

- Left Sidebar (Web Engagement Data From Mynta):** This section contains a search bar and a list of data fields categorized by 'Web Engagement Data From Mynta'. The fields are organized into groups, such as 'Contact Point Email', 'Contact Point Social', 'Contact Point Phone', 'Individual', and 'Web Engagement Data From Mynta'.
- Central Workspace:** This area shows a diagram of the data model. It features nodes representing entities and relationships between them. The nodes are connected by lines, indicating the structure of the data model.
- Right Sidebar (Data Model entities):** This section lists the entities and their attributes. It includes a search bar and a list of entities, each with a set of attributes. The entities are:
  - Contact Point Email:** Attributes include 'Email Address' and 'Email Address'.
  - Contact Point Social:** Attributes include 'Social Media' and 'Social Media'.
  - Contact Point Phone:** Attributes include 'Phone Number' and 'Phone Number'.
  - Individual:** Attributes include 'First Name', 'Last Name', 'Email Address', 'Phone Number', 'Social Media', and 'Web Engagement Data From Mynta'.
  - Web Engagement Data From Mynta:** Attributes include 'Session ID', 'Session Time', 'Page View', 'Click', 'Scroll', 'Hover', and 'Click'.

## Design Model Design End-to-end Salesforce Data Cloud and Agentforce Implementation for Custom 'Einstein Customer Insights Dashboard' LWC



### 4. Calculated Insights Core Configuration

To deliver immediate, operationalized analytics to the LWC search interface, three distinct Calculated Insights (CI) query modules were compiled using SQL aggregate statement architectures:

- **Calculated Insight 1 (Traffic Segmentation Split):** Determines total site activity weights by calculating average engagement score bounds alongside hard tracking counts of active unique sessions, partitioned by user state (Known vs. Unknown). This aggregate analysis isolates user volume differences originating from anonymous traffic versus authenticated shoppers.
- **Calculated Insight 2 (Unified Individual Engagement):** Rolls up and consolidates multi-device event footprints back to a centralized Unified Individual profile identity group. It summarizes total cart additions, wishlists, and order histories to present an un-mocked cross-channel customer health tracker.
- **Calculated Insight 3 (Customer Intent Category Persona Matrix):** Evaluates real-time tracking behavior and cart status to programmatically assign unified individuals into specific marketing segments, determining user engagement thresholds across the storefront canvas.

### 5. Algorithmic Processing & Performance Matrix Reviews

#### A. System Data Normalization Bounds

To protect the reliability of calculated behavioral metrics and lock output score values tightly within a standard 0 to 100 proportional scale, incoming data rows are bounded to operational ranges:

- **Storefront Click Frequencies:** Normalized across a standard range of 0 to 100 clicks.
- **Wishlist Allocation Sizes:** Expected parameter counts capped between 0 to 10 items.
- **Active Shopping Basket Items:** Operational check boundaries restricted to 0 to 10 items in cart.
- **Historical CRM Purchases:** Customer lifecycle transaction tracking counters mapped between 0 to 10 orders.
- **Session Browsing Durations:** Interaction intervals normalized between 0 to 60 minutes.
- **Log Recency Metric Tracks:** Interaction freshness boundaries mapped linearly across a 0 to 30 days window.

B. Engagement Score Proportional Weighting Matrix

|   | A                                       | B          | C      | D                                                                                                                                                       |
|---|-----------------------------------------|------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Storefront Metric Track                 | Max Cap    | Weight | Core Strategic Weightage Rationale                                                                                                                      |
| 2 | CartAdditions (Cart)                    | 10 Items   | 25%    | Highest Intent: Incrementing items into an active checkout basket represents the strongest behavioral proxy of immediate purchase readiness.            |
| 3 | Past Purchase Orders (Orders)           | 10 Orders  | 20%    | Loyalty Factor: Historic fulfillment data proves brand trust; returning buyers demonstrate a structurally higher conversion baseline.                   |
| 4 | WishlistAdditions (Wishlist)            | 10 Items   | 15%    | Future Intent: Documents specific affinity for targeted catalog segments, serving as an ideal anchor for context-aware promotions.                      |
| 5 | Session Browsing Duration (Time)        | 60 Mins    | 15%    | Brand Stickiness: Measures sustained attention spans and deep exploration behavior over simple intermittent page hit counts.                            |
| 6 | Interaction Recency Freshness (Recency) | 30 Days    | 15%    | Freshness Decay: Traffic logged within the last 24 hours defines a critical high-priority lead; score decays lineally as days extend.                   |
| 7 | Click Volume Frequency (Clicks)         | 100 Clicks | 10%    | General Activity: General activity metric that tracks site visibility but can occasionally capture UI navigation confusion if decoupled from cart adds. |

C. Engagement Score Algorithmic Modeling Formula

Standardized Insight Formula Matrix

EngagementScore = 25 \* (Cart / 10) + 20 \* (Orders / 10) + 15 \* (Wishlist / 10) + 15 \* (Time / 60) + 15 \* (1 - Recency / 30) + 10 \* (Clicks / 100)

D. Customer Intent Core Calculation Modeling

To translate quantitative scoring records into explicit, context-aware service advice blocks, the application computes user intent personas by passing engagement trends and CRM membership parameters into a nested conditional statement:

Logical Customer Intent Selection Formula

```
=IF(EngagementScore >= 70, "High Intent: Purchase Ready",
IF(AND(EngagementScore >= 40, Loyalty = ""), "High Intent: New Lead",
IF(AND(EngagementScore >= 40, Loyalty <> ""), "Mid Intent: Retention",
IF(EngagementScore >= 25, "Low Intent: Researcher", "Passive: Window Shopper"))))
```

Strategic Segmentation Rationale & Logic Breakdown:

- **High Intent (Purchase Ready):** Any individual scoring an aggregate value >= 70 is flagged as an absolute outreach priority, completely bypassing loyalty filters. At this phase, concentrated checkout cart activity and high browse frequencies represent a customer who is ready to buy.
- **High Intent (New Lead):** Triggered when high engagement bounds (>= 40) match an unauthenticated or blank CRM Loyalty status field (Loyalty = ""). This highlights a high-value anonymous user exploring the catalog—defining the optimal window to capture an email or present a first-time account setup discount.
- **Mid Intent (Retention):** Fired when a strong activity score (>= 40) is paired with an active, populated loyalty database entry (Loyalty <> "", e.g. Platinum, Gold fans). This identifies returning core advocates who respond best to loyalty milestones, club points rewards, or exclusive product access updates.
- **Low Intent / Researcher / Window Shopper:** Characterized by lower performance tiers (25 to 39 for Researchers, and < 25 for passive Window Shoppers). This indicates early-stage product exploration, catalog browsing, or general window shopping profiles where active checkout intent is absent.

E. System Architecture Execution Review Matrix

|   | A                                                                                                                                                                                                  | B                                                                                                                                                                                           | C                                                                                                                                                                                                       | D                                                                                                                                                                            |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Design Decisions                                                                                                                                                                                   | Technical Challenges Faced                                                                                                                                                                  | Strategic Trade-offs Considered                                                                                                                                                                         | Future Roadmap Enhancements                                                                                                                                                  |
| 2 | ConnectApi High-Scale Data Queries:<br>Replaced traditional inline SOQL statements with ConnectApi.CdpQuery parameters to query virtualized tables cleanly without core engine compilation blocks. | Transform DMO Retrieval Bugs:<br>Bypassed the native platform search index visibility limitation by mapping static transaction sets straight into a custom schema table container.          | Client-Side UI Regular Expression Parsing:<br>Chose client-side regex manipulation in LWC over altering backend setups. This isolates recommendation phrases quickly while preserving template agility. | Identity Resolution Auto-Linking:<br>Extend match criteria rules to automatically link historic anonymous user cookies with target customer accounts immediately post-login. |
| 3 | Absolute Unit Typings:<br>Eliminated fluid system token references in the CSS file to prevent organizational token access failures, ensuring long-term rendering stability.                        | Precision Trailing Decimal Floats:<br>Managed high-precision trailing data pads from external tables (e.g. phone fields) using Apex string truncation parameters like substringBefore('.'). | Lightweight Iterative Markup Loops:<br>Used simple HTML layouts and responsive iteration elements over heavy grid components to speed up dashboard render loops.                                        | Dynamic Wire Serialization Caching:<br>Incorporate client-side data serialization blocks to minimize redundant sever-side query lookups during navigation transitions.       |

## 6. Agentforce Advisory Configuration & Template Guidance

The top orchestration layer defines an autonomous Agentforce Advisory Agent equipped with intent-focused subagents specialized in real-time customer context evaluation. When an employee searches a visitor ID inside the custom dashboard, the subagent analyzes current tracking parameters and triggers the grounded recommendation action sequence. This subagent calls the custom Einstein Prompt Template Advisory\_Prompt, passing the string variable Input:Customer\_Name to anchor generation parameters. The template uses fine-tuning prompt guidelines to instruct the Einstein LLM to query the registered vector retriever, look up the consolidated customer footprint data, parse current Calculated Insights matrices, and isolate a targeted, next-best-action text string block without generic background boilerplate padding.